
Ranking Free RAG: Replacing Re-ranking with Selection in RAG for Sensitive Domains

Yash Saxena¹ Ankur Padia¹ Mandar S. Chaudhary^{2†} Kalpa Gunaratna³ Srinivasan Parthasarathy⁴
Manas Gaur¹

Abstract

Retrieval-Augmented Generation (RAG) systems deployed in sensitive domains must provide interpretable evidence selection and robust safeguards against data poisoning, yet current approaches rely on opaque similarity-based retrieval with arbitrary top- k cutoffs that offer no explanation for their selections and remain vulnerable to adversarial manipulation. METEORA replaces re-ranking with rationale-driven selection via three components: a DPO-tuned LLM that generates explicit retrieval rationales, an Evidence Chunk Selection Engine (ECSE) that uses those rationales with statistical elbow detection for adaptive cutoff determination, and a Verifier LLM that filters poisoned evidence using the same rationales. Across six datasets, METEORA achieves 13.41% higher recall, 21.05% higher precision (without expansion), an 80% reduction in evidence volume, a 33.34% improvement in answer accuracy, and a 4.4 $\times$ improvement in adversarial robustness. Human evaluation confirms genuine interpretability (3.64/5 confidence; 86% ground-truth agreement), demonstrating that interpretability, efficiency, and robustness are synergistic rather than competing objectives. The code is available in the GitHub repository <https://github.com/YashSaxena21/METEORA>

1. Introduction

Traditional Retrieval-Augmented Generation (RAG) systems retrieve and re-rank thousands of document evidences

[†] This work does not relate to the author’s position at eBay Inc. ¹University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA ²eBay Inc., San Jose, California, USA ³Independent Researcher ⁴Ohio State University, Columbus, Ohio, USA. Correspondence to: Yash Saxena <ysaxena1@umbc.edu>, Manas Gaur <manas@umbc.edu>.

to answer queries, mostly focusing on improving accuracy. They provide no explanation for *why specific evidence* was chosen over alternatives. This opacity creates critical problems in sensitive domains where understanding the reasoning behind AI decisions is not optional; it is essential for auditability and transparency. Consider a legal AI system analyzing merger agreements or a healthcare platform processing patient data. When these systems select certain evidence chunks to generate responses, stakeholders need to understand the selection rationale. Current similarity-based re-ranking methods provide only opaque similarity scores and rely on arbitrary top- k cutoffs, offering no insight into the decision-making process (Glass et al., 2022; Reimers and Gurevych, 2019a; Devlin et al., 2019). This black box nature has become untenable as organizations deploy RAG in high-stakes scenarios where wrong answers can have serious regulatory consequences (Zhou et al., 2024a; Barron et al., 2024).

The interpretability crisis becomes even more critical when considering adversarial threats. Recent demonstrations have shown how attackers can poison RAG knowledge bases to manipulate system outputs (Zou et al., 2025; Nazary et al., 2025), but current re-ranking approaches provide no mechanism to detect such manipulation. Without understanding *why evidence was selected*, organizations cannot audit whether the selection process has been compromised. The same opacity that prevents explainability also prevents robust defense against corpus poisoning attacks. In high-stakes domains, these adversarial attacks can have serious consequences. These attacks exploit the black box nature of evidence selection: if the system cannot explain its reasoning, it cannot detect when that reasoning has been subverted.

Current approaches miss the connection between interpretability and robustness. RAG² (Sohn et al., 2025a) and RADIO (Jia et al., 2025) improve retrieval through rationales but still rely on traditional re-ranking, inheriting its opacity. RankRAG (Yu et al., 2024a) uses Large Language Models (LLMs) for ranking but provides no reasoning transparency and lacks mechanisms to detect adversarial manipulation. Figure 1 shows that these methods treat interpretability and security as separate concerns: traditional re-rankers select irrelevant and poisoned evidence based on

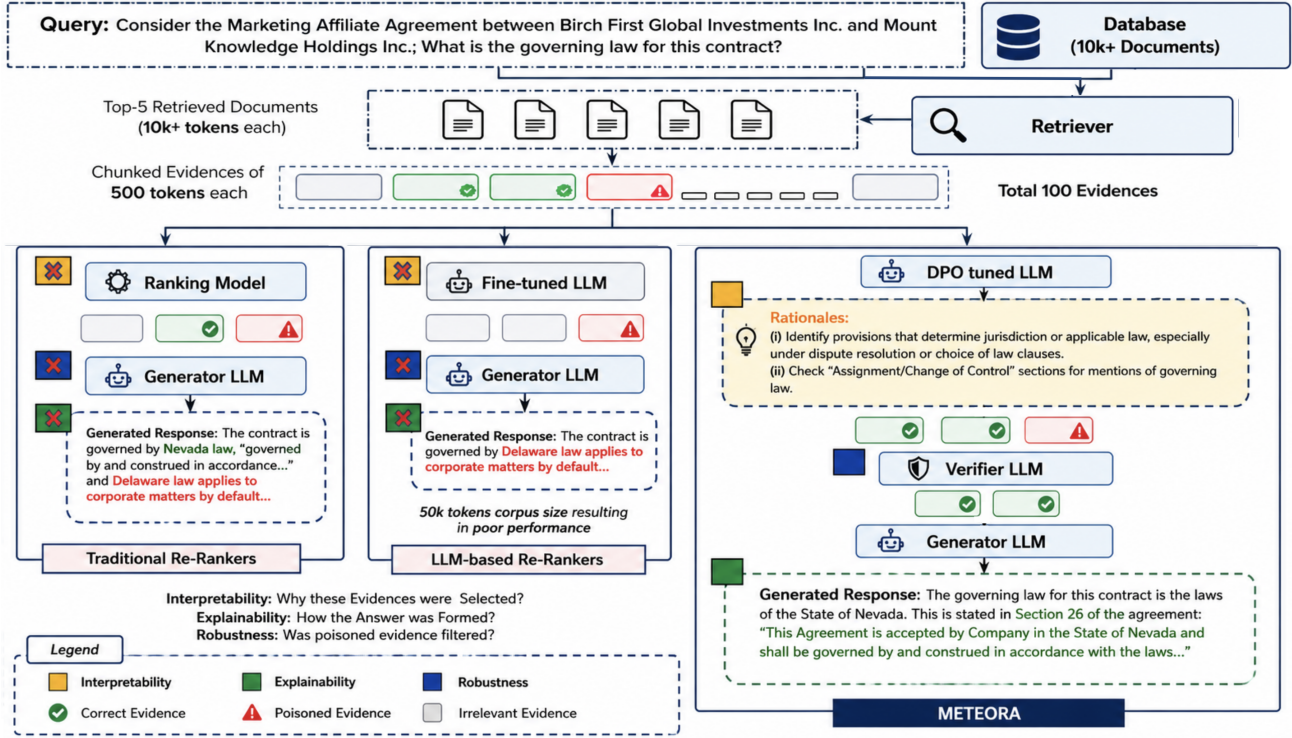


Figure 1. METEORA achieves interpretable and robust evidence selection where existing approaches fail. On a challenging legal query from the Merger Agreement Understanding Dataset with poisoned evidence, traditional re-rankers select contaminated content producing factually incorrect answers, while LLM-based re-rankers fail due to context length limitations. **Only METEORA uses interpretable rationales to explain evidence selection and robustness with verification to filter poisoned content, achieving both interpretability and correctness.**

opaque similarity scores, while LLM-based re-rankers fail when context length exceeds their capacity, both lacking mechanisms to explain or verify their selection decisions.

We propose METEORA (**M**ethod for **I**nterpretable rank-free evidence selection with **O**ptimal **R**ationale), a framework that solves the black box problem by replacing opaque re-ranking with explicit reasoning. As demonstrated in Figure 1, METEORA generates rationales that explain *why* specific evidence is relevant to a query, then uses these same rationales to verify evidence consistency and detect adversarial content. Unlike LLM-based re-rankers that fail with long documents due to context length limitations, METEORA operates without such constraints while providing a verifier that successfully filters poisoned evidence. Crucially, this enhanced interpretability and robustness come *without sacrificing efficiency*, METEORA’s adaptive selection actually reduces computational overhead by processing only necessary evidence rather than arbitrary top- k evidence.

Our key insight is that interpretable evidence selection, adversarial robustness, and computational efficiency are not competing goals but synergistic ones. By making the selection process explainable through rationales, we enable both human understanding and automated verification while simultaneously eliminating the computational waste inher-

ent in arbitrary top- k approaches. The same reasoning that justifies evidence selection can flag inconsistencies that indicate adversarial manipulation, and the adaptive nature of rationale-driven selection naturally processes only the evidence needed for accurate responses.

Our main contributions are:

- **DPO-based rationale generation framework:** To address the fundamental lack of explainability in RAG evidence selection, we introduce a novel approach for training rationale generators using Direct Preference Optimization (DPO) without manual annotation, enabling scalable production of query-aligned reasoning that makes evidence selection auditable and transparent in sensitive domains.
- **Evidence Chunk Selection Engine (ECSE) with statistical adaptive thresholding:** To eliminate the arbitrary and computationally wasteful nature of top- k heuristics that process irrelevant evidence in current re-ranking methods, we develop an unsupervised evidence selection algorithm using statistical elbow detection with z-score normalization, providing principled, data-driven cutoff determination that achieves both computational efficiency and query-adaptive precision.
- **Unified rationale-based selection and verification:** To

solve the fundamental disconnect between interpretability and adversarial robustness in current RAG systems, we establish a methodological framework where the same rationales that explain evidence selection also enable detection of corpus poisoning attacks, proving that transparency and security are not competing goals but synergistic properties of reasoning-based selection.

2. Related Work

RAG in Sensitive Domains. RAG has been widely adopted in high-stakes domains such as law, finance, and healthcare, where factual accuracy and verifiability are critical (Lewis et al., 2020; Li et al., 2025; Sohn et al., 2025b; Gyöngyössy et al., 2024; Bhushan et al., 2025). Such domains are regulated and hence require traceable generations to retrieve and select sources (for example, denial of loan application) and are prone to select semantically similar but contextually misleading evidence (Gyöngyössy et al., 2024; Bhushan et al., 2025). Furthermore, RAG pipelines remain susceptible to corpus-poisoning attacks (Broomfield et al., 2025; Zhou et al., 2024b), highlighting the critical need for secure, context-aware retrieval methods. Our framework addresses these vulnerabilities by producing query-specific rationales that explain and justify chunk/evidence selection, thereby enhancing the system’s transparency, interpretability, and overall robustness.

Heuristics in RAG. Most RAG systems use fixed top- k selection heuristics, which often hurt performance due to irrelevant or noisy context (Cheng et al., 2025; Asai et al., 2024; Gaur et al., 2022). Moreover, in real-world applications, it is hard to know upfront what the value should be for k , and re-rankers often lack interpretability, which hence complicates their deployment in sensitive domains. While some efforts have been made to automate the selection of k (Chen et al., 2018; Ren et al., 2024), these methods have had limited success due to their inability to explain the selection of evidence to generate final text. Dynamic retrieval methods such as RankRAG (Yu et al., 2024b) and Self-RAG (Asai et al., 2024) improve adaptability but lack interpretability. In contrast, our METEORA replaces top- k heuristics with rationale-grounded selection, enabling query-specific and explainable document filtering.

Interpretability in RAG. Interpretability is often absent in RAG pipelines. Recent efforts such as MIRAGE (Qi et al., 2024) and Rationale-first Retrieval (Sohn et al., 2025b) introduce rationales or attribution signals, but they focus on improving generation faithfulness and retriever training, respectively. In METEORA, we target the re-ranking stage and define end-to-end rationale integration by using rationales both to select evidence and to verify them before generation. Set-R (Lee et al., 2025) also employs rationales during re-ranking, but it continues to use LLM-based

ranking and does not address adversarial robustness, which limits ranking and verification effectiveness compared with METEORA. Unlike Sohn et al. (2025b), which still relies on downstream re-ranking and lacks verification, METEORA removes re-ranking entirely and applies rationale-derived instructions in both selection and filtering, yielding interpretable decisions.

Reliability in RAG. RAG systems are susceptible to adversarial attacks and retrieval of noisy evidence (Broomfield et al., 2025; Xue et al., 2024). Methods like entailment filtering (Cheng et al., 2025) and ensemble retrieval offer partial solutions. These approaches address only specific vulnerabilities while leaving others unaddressed - entailment filtering can be overly strict and discard valuable information that doesn’t meet formal entailment criteria, while ensemble methods add complexity and computational overhead without fundamentally solving the semantic verification problem. On the other hand, METEORA adds semantic filtering by requiring coherent rationales per evidence, to help discard poisoned evidence before generation.

Feedback-Based Optimization. Policy optimization methods like PPO (Ouyang et al., 2022), RLHF, and GRPO (Shao et al., 2024) have been used to align LLMs with human preferences, but rely on complex reward models and unstable training dynamics. In contrast, DPO (Rafailov et al., 2023a) offers a simpler, supervised alternative that directly optimizes preferred outputs. While prior methods focus on generation helpfulness or safety, we demonstrate that DPO is well-suited for rationale generation, where precision and domain adherence are crucial. PPO and RLHF suffer from costly reward modeling, opaque optimization processes, and vulnerability to data poisoning, whereas GRPO reduces but doesn’t eliminate these issues. DPO overcomes these limitations by providing explicit, interpretable reasoning paths for evidence selection, making the rationale generation process not only more transparent but also more robust against poisoned content. We apply DPO in METEORA to produce rationales that align with domain-specific expectations while avoiding the pitfalls of traditional policy gradients.

3. Proposed Framework

Problem Formulation. Formally, given a query q , knowledge base D , and retrieved documents $D_s \subset D$ chunked into evidence set E , the objective is to learn a rationale-based selection function $f_\theta(q, E) \rightarrow (R, E_s)$ that generates rationales $R = \{r_1, \dots, r_k\}$ and selects evidence subset $E_s \subset E$ such that: (i) E_s maximizes recall $R(E_s, E^*)$ where E^* are ground-truth relevant evidences, (ii) E_s minimizes precision loss from adversarial content $\mathcal{A} \subset E$, (iii) $|E_s|$ is determined adaptively without fixed top- k constraints, and (iv) rationales R provide interpretable justification for each $e \in E_s$. Figure 1 shows an example of a

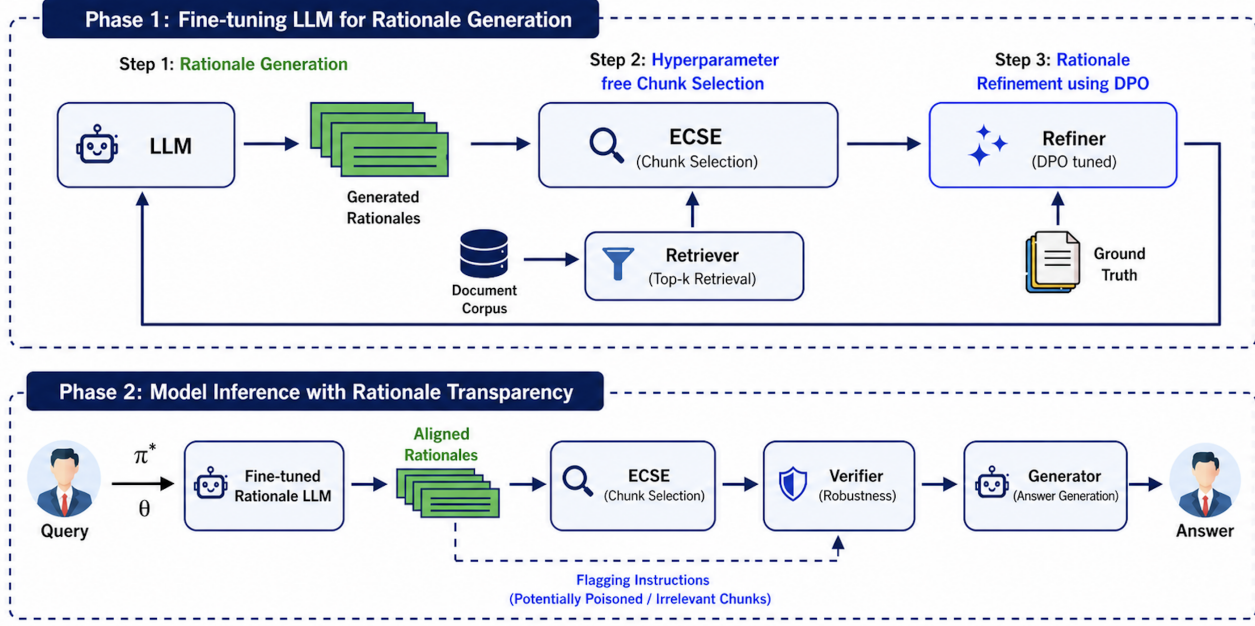


Figure 2. Overview of our METEORA framework.

rationale generated for a real-world query from the PrivacyQA dataset. As shown in Figure 2, METEORA comprises three major components: the preference-tuned rationale generator, the ECSE, and the verifier. We describe each component in the following subsections.

3.1. Preference-Tuned Rationale Generator

METEORA’s rationale generator produces query-aligned reasoning that explains evidence selection decisions. Rather than relying on manual annotation for rationale creation (Wei et al., 2022; Kim et al., 2023; Zaidan et al., 2007), we use Direct Preference Optimization (DPO) to automatically train a general-purpose LLM to generate rationales that lead to correct evidence selection. DPO offers significant advantages over traditional RLHF approaches: it eliminates the need for reward model training and complex reinforcement learning pipelines, provides more stable training through simple classification loss, and requires substantially lower computational resources (Rafailov et al., 2023b).

Preference Dataset Construction. We automatically construct preference datasets from existing QA annotations without manual labeling. For each query q and ground-truth evidence e^* , we generate multiple rationales and label those that lead to correct evidence selection as preferred (r_w) and those that select incorrect evidence as dispreferred (r_l). This approach leverages the insight that rationale quality can be measured by selection accuracy rather than human judgment.

DPO Training Objective. We train the rationale generator

using the DPO loss function ($\mathcal{L}_{DPO}(\pi_\theta; \pi_{\text{ref}})$):

$$= -\mathbb{E}_{(q, e, r_w, r_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(r_w | q, e)}{\pi_{\text{ref}}(r_w | q, e)} - \beta \log \frac{\pi_\theta(r_l | q, e)}{\pi_{\text{ref}}(r_l | q, e)} \right) \right]$$

where π_{ref} is the frozen reference model, π_θ is the trainable model with the same architecture, and β controls preference learning intensity. The model learns to assign higher likelihood to rationales r_w that lead to correct evidence selection over rationales r_l that select incorrect evidence, given query q and evidence e .

Training vs. Inference. During training, we condition on query and ground-truth evidence: $\pi_\theta(r | q, e^*)$, to enable the model to learn what good rationales look like when paired with correct evidence. At inference time, we condition only on the query: $\pi_\theta(r | q)$, to allow the model to generate query-appropriate rationales to guide evidence selection. This training paradigm enables the model to internalize effective reasoning patterns and apply them to new queries, to achieve superior performance compared to baselines (section 5). The resulting preference-tuned model π_θ^* generates query-aligned rationales to serve dual purposes in METEORA: guide evidence selection through the Evidence Chunk Selection Engine (subsection 3.2) and enable adversarial detection through the Verifier LLM (subsection 3.3). Implementation details are provided in Table 6, with prompts in §A.1 and qualitative examples in §A.2.

Note on Adversarial Robustness. The preference-tuned

rationale generator provides implicit defense against coincidental selection of adversarial evidence through three mechanisms: (1) specificity learning, where DPO training optimizes rationales for fine-grained features that distinguish correct from incorrect evidence, making coincidental matching with adversarial content less likely; (2) query-evidence alignment, where conditioning on $\pi_\theta(r|q, e^*)$ learns specific relational patterns that adversarial content would need to coincidentally match; and (3) selection accuracy pressure, where the preference objective reinforces rationales robust to superficial similarities. However, this implicit defense has limitations; negative examples r_l represent naturally incorrect evidence rather than adversarially crafted content, creating a training distribution gap. While rationale specificity helps, adversarial evidence is primarily handled by the Verifier LLM (subsection 3.3), which checks rationale consistency to flag inconsistent or malicious content.

3.2. Evidence Chunk Selection Engine

The Evidence Chunk Selection Engine (ECSE) is the unsupervised component that uses generated rationales to **eliminate top- k heuristics** to adaptively decide the number of evidence chunks to select. ECSE uses two complementary mechanisms to collectively identify relevant evidence without using arbitrary cutoffs.

Stage 1: Pairing Rationales with Evidence. The DPO-tuned LLM generates *locally relevant, query-specific rationales*, meaning each rationale is tailored to and depends solely on the individual query being processed. To capture this query-specific trait, we perform rationale-based pairing. Each rationale captures a specific aspect of what makes evidence relevant to the query. To leverage this specificity, we pair each rationale $r_i \in \mathcal{R}$ with its most semantically similar evidence chunk, forming the set $E_v = \{ \arg \max_{e_j \in E} \mathcal{S}(r_i, e_j) \mid r_i \in \mathcal{R} \}$, where $\mathcal{S}(\cdot, \cdot)$ denotes cosine similarity.

Note on Rationale Convergence: Multiple rationales may select the same evidence chunk when they capture complementary aspects of the same relevant content, resulting in $|E_v| \leq |\mathcal{R}|$. This convergence is desirable as it indicates consensus among rationales about highly relevant evidence. Rather than forcing diversity, we interpret this as evidence validation. When multiple reasoning paths point to the same content, it strengthens confidence in that evidence’s relevance. This pairing ensures that each rationale’s specific intent is represented in the selected evidence, providing high precision through rationale-evidence alignment.

Dataset-Specific Relevance: Further, to capture the dataset-specific trait, we identify evidence that align with the overall intent of all rationales. To identify such evidence, we compute a pooled rationale embedding: $\bar{r} = \frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \text{SBERT}(r_i)$.

We rank all evidence chunks by similarity to $\bar{r}$, yielding similarity $\{s_1, s_2, \dots, s_n\}$ in descending order. To determine the adaptive cutoff k^* without top- k heuristics, we apply statistical elbow detection: (a) Compute first-order differences: $\Delta_i = s_i - s_{i+1}$; (b) Apply z-score normalization: $z_i = \frac{\Delta_i - \mu_\Delta}{\sigma_\Delta}$ where μ_Δ and σ_Δ are the mean and standard deviation of all differences. (c) The selection index k^* is identified at the first point where the drop in similarity significantly deviates from the average pattern, indicating a natural boundary between highly relevant and less relevant evidence.

This identifies the point where similarity drops significantly, indicating a natural boundary between relevant and irrelevant evidence. If no significant drop is detected (all $z_i \leq \tau$), we use second-order differences $\nabla_i^2 = \Delta_{i+1} - \Delta_i$ to find maximum curvature: $k^* = \arg \max_i |\nabla_i^2|$. The selected evidence set is $E_g = \{e_1, e_2, \dots, e_{k^*}\}$.

Stage 2: Context Expansion (Addressing Chunking Brittleness; optional). Document chunking can artificially split coherent information across adjacent evidence. To recover potentially fragmented context, we expand each selected evidence in $E_v \cup E_g$ by including its immediate neighbors:

$$E_w = \{e_{j-1}, e_{j+1} \mid e_j \in (E_v \cup E_g), j-1 \geq 1, j+1 \leq |E|\}$$

The final evidence set is $E_s = E_v \cup E_g \cup E_w$. Note that this step is optional (Stabler et al., 2025); if the chunking process is sufficiently robust. The expansion involves a precision-recall trade-off: while it recovers split information, it may introduce irrelevant content when evidence is not actually fragmented.

Computational Complexity Analysis. To analyze METEORA’s efficiency, we compare the computational costs of traditional RAG against our approach using standard big-O notation, where n represents the number of retrieved evidences, r the number of rationales, and d the embedding dimension. Traditional RAG systems follow a straightforward three-step process. Re-ranking requires $O(n \cdot d)$ operations to compute similarity scores between the query and all n evidence. Top- k selection then sorts these evidences in $O(n \log n)$ time to identify the most relevant ones. Finally, generation processes the selected k evidence through the LLM at cost $O(k \cdot L_{\text{prefill}})$, where L_{prefill} represents the computational expense of LLM processing per evidence.

METEORA’s pipeline involves more steps but achieves better overall efficiency. Rationale generation requires $O(L_{\text{rat}})$ for a single LLM call. ECSE pairing then computes similarities between r rationales and all n evidences at cost $O(r \cdot n \cdot d)$. The pooling stage combines rationale embeddings in $O(r \cdot d)$ time, computes evidence similarities in $O(n \cdot d)$, sorts results in $O(n \log n)$, and performs z-score elbow detection in $O(n)$, totaling $O(r \cdot d + n \cdot d + n \log n + n)$. Context expansion adds $O(|E_v \cup E_g|)$ operations for neighbor lookup.

Table 1. Empirical latency comparison under the same hardware and model setup. TTFT denotes time to first token, and Total Time = TTFT + Generation Time.

Method	Input Tokens	TTFT (s)	Gen. Time (s)	Total (s)
SBERT	36.89k	3.07	0.97	4.04
RankRAG	39.82k	3.52	1.09	4.61
METEORA w/o Verifier	18.68k	1.21	0.56	1.77
METEORA	12.23k	2.42	0.49	2.91

Finally, generation processes only k^* adaptively selected evidences at cost $O(k^* \cdot L_{\text{prefill}})$. METEORA selects about **~80% fewer evidences** than the traditional top- k baseline for achieving similar recall, that is $k^* \ll k$. With short generations, $L_{\text{prefill}} \gg L_{\text{rat}}$, so prefill dominates total cost. Since both methods produce outputs of similar length, reducing the context size directly reduces compute, yielding substantial net savings for METEORA.

Empirical Latency Analysis. To quantify the practical cost of rationale generation and verification, we measured latency under the same hardware and model setup for all methods. We report input tokens, time to first token (TTFT), generation time, and total time, where Total Time = TTFT + Generation Time. As shown in Table 1, METEORA w/o Verifier takes 1.77s, while full METEORA takes 2.91s, so the verifier adds approximately 1.14s per query. Even with this additional verification step, full METEORA remains faster than SBERT re-ranking (4.04s) and RankRAG (4.61s) because it processes substantially fewer input tokens overall. Thus, rationale generation and verification introduce sequential computation, but the reduction in evidence volume more than compensates for this cost in practice.

3.3. Verifier LLM

To enhance METEORA’s robustness against contradictions and factual inconsistencies, we incorporate a Verifier LLM to perform conservative consistency checking on selected evidence E_s before generation. The Verifier uses the same rationale framework to guide evidence selection, providing a secondary filtering mechanism within METEORA’s unified reasoning approach.

Verification Process. The Verifier evaluates each evidence using the input query, associated rationales (which serve as flagging instructions), and summaries of previously verified evidence. Following a conservative design, the Verifier assumes validity unless strong evidence suggests otherwise and only flags content when highly confident ($> 90\%$). Evidence is flagged for three types of issues: (1) *factual violations*, when content contradicts well-established facts with high confidence; (2) *contradiction*, when evidence is logically inconsistent with previously verified evidences; and (3) *instruction violations*, when evidence fails to meet criteria embedded in the rationales (Greshake et al., 2023; Yi et al., 2025; Zou et al., 2025; Clop and Teglia, 2024; Ben-

Tov and Sharif, 2025; Parry et al., 2024). Flagged evidence is discarded, and the filtered set proceeds to generation.

We use the same Llama-3.1-8b-instruct model for verification to maintain consistency across the framework. The Verifier processes each piece of evidence independently and outputs structured decisions, including flagging status, evidence summaries, and violation types. This conservative approach prioritizes precision over recall in adversarial detection, reducing false positives that could harm system performance on clean data. For more information, see §A.1.

4. Experiments

To demonstrate the effectiveness of rationales, we evaluate METEORA on three tasks across six real-world benchmark datasets spanning multiple domains, and provide a detailed ablation study. For clarity, we report results using evidence of chunk size of 512 tokens in the main paper; results for other sizes are included in §A.5.

Tasks and Datasets: We evaluate METEORA’s core properties with three complementary tasks to measure different aspects of rationale-driven evidence selection.

(a) *Context Prioritization (CP)* measures METEORA’s fundamental capability to eliminate top- k heuristics through adaptive evidence selection. This task evaluates precision, recall, and F1 scores against human-annotated ground truth evidence spans, directly testing whether statistical elbow detection identifies relevant content more accurately than arbitrary cutoffs. We used LLaMA-3.1-8b-Instruct for rationale generation, evidence verification, and answer generation. Baseline re-ranking methods varied top- k from 1-64 across datasets and evidence sizes. Since METEORA doesn’t use fixed k -values, for fair comparison, we set each baseline’s k -value to match the average number of evidences selected by METEORA and measure performance using Precision@ k and Recall@ k .

(b) *Generation Quality* assesses whether improved evidence selection translates to better answer accuracy. By comparing generated responses against reference answers, this task validates that selecting fewer but more relevant evidence enhances downstream performance, the core efficiency claim of adaptive selection. We measure generation quality using GPT-4o as a binary evaluator (+1 correct, 0 incorrect).

(c) *Adversarial Defense* Adversarial Defense: Following (Zou et al., 2025)’s protocol, we simulated corpus poisoning using domain-specific LLMs to generate semantically coherent but factually incorrect content. We used Law-Chat for legal datasets, Finma-7b-full for financial data, and LLaMA-3.1-8b-Instruct for academic content. We randomly poisoned 30% of QA instances by inserting malicious text into documents containing correct context. This

Table 2. Dataset Statistics Across Domains

Dataset	Docs	Avg. Tokens	QA Pairs	Domain
ContractNLI	95	10,673	946	Legal (NDA)
CUAD	462	55,827	4,042	Legal (Contracts)
MAUD	150	351,476	1,676	Legal (M&A)
PrivacyQA	7	25,266	194	Legal (Privacy)
FinQA	2,789	~700	8,281	Finance
QASPER	1,585	~6,500	5,049	Academic (NLP)

creates a particularly challenging detection scenario since poisoned content appears alongside legitimate information in the same documents, making it difficult to distinguish based on location or semantic similarity alone. Adversarial robustness was measured using precision and recall for detecting poisoned content.

Dataset Selection Criteria. We selected six datasets spanning legal, financial, and academic domains based on three evaluation requirements: (1) expert-annotated evidence spans for precise CP evaluation, (2) lengthy documents (5-50 pages) that stress-test adaptive selection under strong evidence counts, and (3) domain diversity to validate generalization without parameter retuning. Legal datasets include ContractNLI (Koreeda and Manning, 2021), PrivacyQA (Ravichander et al., 2019), CUAD (Hendrycks et al., 2021), and MAUD (Pipitone and Alami, 2024), representing increasing complexity from straightforward privacy policies to highly technical merger and acquisition agreements (M&A). FinQA provides numerical reasoning challenges over financial reports with hybrid text-table evidence (Chen et al., 2021). QASPER covers academic research papers where questions target specific empirical claims requiring precise evidence selection (Dasigi et al., 2021). Table 2 shows the statistics. Such an evaluation setup directly tests METEORA’s core contributions: interpretable evidence selection through explicit rationales, principled elimination of arbitrary top- k heuristics via adaptive thresholding, and robustness against adversarial content with rationale-based verification.

Baselines. We evaluate against established re-ranking methods representing different paradigms in retrieval systems. For context prioritization, we compare with Cross-Encoder (ms-marco-MiniLM-L4-v2, bge-reranker-large) (Huggingface), which processes query-document pairs jointly to produce fine-grained relevance scores. We also test against Contriever (Izacard et al., 2022), an unsupervised dense retriever that learns representations through contrastive learning, and SBERT (ms-marco-MiniLM-L4-v2, e5-large-v2) (Reimers and Gurevych, 2019b) to compute similarity between independent query and document embeddings. Additionally, we include Fine-Tuned SBERT (Legal-huggingface), a domain-adapted version specialized for legal text, to assess the impact of domain-specific training. For LLM-based comparison, we use LLM-based rerankers such as RankLLaMa (Ma et al., 2024) and implement state-of-the-art RankRAG

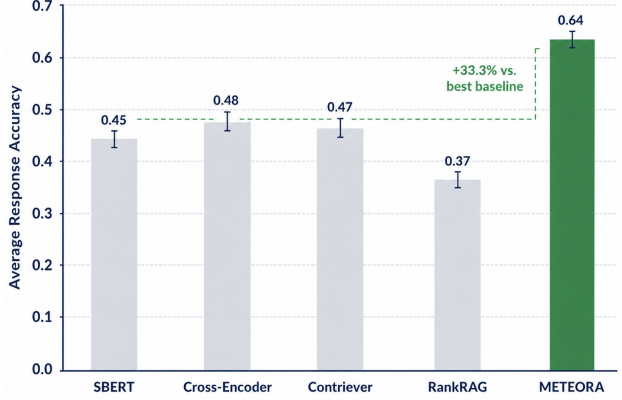


Figure 3. Response accuracy when retrieved evidence matches ground-truth, with variance bars indicating statistical significance ($p < 0.01$, paired t-test) across trials.

using the Promptriever model (Weller et al., 2024), which leverages large language models to directly rank and select relevant evidence. Finally, we evaluate adversarial robustness against Perplexity-based Defense (Zhou et al., 2024b), which detects poisoned content by measuring deviations from expected language model behavior.

5. Results

Results from CP and Generation Tasks. Table 3 reports performance on the CP task across all datasets; our baselines include six traditional re-rankers and two LLM-based re-rankers; the METEORA framework not only provides interpretability (see qualitative examples in §A.2) but also surpasses state-of-the-art baselines in retrieval performance. On individual datasets, METEORA improves recall over the best-performing baseline by 5.31% (QASPER), 8.6% (Contract-NLI), 10.11% (PrivacyQA), 19.23% (CUAD), and 41.17% (MAUD); these results indicate that as dataset complexity and average document length increase from 6.5k to 350k tokens (see Table 2), the advantage of METEORA becomes more pronounced. METEORA’s advantage grows as datasets become more challenging, precisely where similarity-based methods hit their ceiling: (a) DPO training first learns to bridge the gap between vague questions and where answers actually reside, aligning intent with evidence-bearing regions, in Scientific QASPER, where questions are written from titles and abstracts while answers lie deep in full papers, and transferring the same behavior to legal corpora, where it maps queries to the specific clauses that support, contradict, or ignore a claim. See appendix §D for theoretical details. (b) Rationale generation then converts this learned intent into precise retrieval targets, turning “How does this model work?” into searches for methodology sections, experimental protocols, and evaluation metrics in scientific papers, and showing the parallel shift in legal datasets from surface keywords in Contract-

Table 3. CP Task Results across Datasets: METEORA achieves the best average recall and precision, outperforming all baselines. Since METEORA does not depend on a specific K value, we established a fair comparison by taking the average number of evidence it selected and using that number as the K value for all other baselines. The metrics are Precision (P@K) and Recall (R@K). Values in **bold** exceed state-of-the-art alternatives.

Model	QASPER		Contract-NLI		FinQA		PrivacyQA		CUAD		MAUD		Average	
	R@8	P@8	R@3	P@3	R@10	P@10	R@6	P@6	R@12	P@12	R@33	P@33	R	P
SBERT (MiniLM-L4-v2)	0.91	0.26	0.91	0.38	0.96	0.12	0.89	0.24	0.78	0.11	0.41	0.03	0.81	0.19
SBERT (E5-Large-v2)	0.94	0.27	0.92	0.34	0.96	0.12	0.78	0.21	0.77	0.12	0.44	0.02	0.80	0.18
Contriever	0.94	0.25	0.89	0.36	0.98	0.10	0.82	0.23	0.73	0.10	0.46	0.01	0.80	0.17
Cross-Encoder (MiniLM-L4-v2)	0.94	0.27	0.91	0.38	0.97	0.10	0.81	0.22	0.71	0.10	0.50	0.02	0.80	0.18
Cross-Encoder (BGE-Reranker-Large)	0.93	0.27	0.92	0.37	0.97	0.11	0.85	0.22	0.78	0.11	0.51	0.02	0.82	0.18
Finetuned-SBERT	0.92	0.26	0.92	0.39	0.97	0.13	0.75	0.21	0.76	0.11	0.46	0.02	0.79	0.18
RankLLaMA (7b)	0.75	0.19	0.73	0.24	0.85	0.08	0.83	0.15	0.71	0.08	0.27	0.02	0.69	0.12
RankRAG (8b)	0.76	0.19	0.77	0.23	0.89	0.07	0.86	0.19	0.60	0.07	0.22	0.01	0.68	0.13
METEORA w/o DPO	0.96	0.25	0.95	0.37	0.92	0.11	0.92	0.22	0.89	0.12	0.65	0.02	0.88	0.18
METEORA w/o Verifier	1.00	0.25	1.00	0.34	0.97	0.10	1.00	0.22	0.94	0.11	0.75	0.02	0.94	0.17
METEORA	0.99	0.26	1.00	0.35	0.95	0.12	0.98	0.23	0.93	0.12	0.72	0.03	0.93	0.19
METEORA w/o Expansion	0.96	0.31	0.97	0.45	0.94	0.14	0.96	0.27	0.90	0.14	0.66	0.04	0.89	0.23

NLI to higher level concepts in CUAD and MAUD, recognizing that “liability limitation,” “damages exclusion,” and “loss indemnification” refer to same concept. (c) Adaptive thresholding finally decides how much evidence to gather, adjusting to uneven information density in both scientific articles and contracts, stopping when sufficient relevant content is found; simple papers or contracts need fewer chunks, complex studies or merger agreements need more, and the statistical detector finds natural breakpoints without fixed cutoffs.

On FinQA, which has relatively short documents (average length ~ 700 tokens), METEORA is not the most effective method in terms of recall because the dataset is constructed from passages that already contain the answer rather than from entire documents. This setup does not reflect real-world scenarios where answers could appear anywhere in long, complex documents, and in this constrained setting, traditional similarity-based re-rankers achieve higher recall (0.98 vs. 0.95). Nevertheless, METEORA achieves a significant reduction in the number of evidences ($\sim 80\%$ fewer evidence, refer to Table 4) compared to strong baselines, cutting noisy context to boost generation accuracy with more grounded responses (see Figure 3).

To examine whether these efficiency gains hold in more complex evidence settings, we further study METEORA under multi-hop conditions. Multi-hop settings are substantially more challenging than single-hop settings, and advanced techniques often struggle under such conditions (Maram et al., 2026). Since multi-hop performance depends heavily on retrieving and preserving all necessary evidence, we examine whether METEORA’s adaptive selection affects single-hop and multi-hop performance differently. We analyze 20K+ instances from QASPER, Contract-NLI, PrivacyQA, CUAD, and MAUD, all of which require **multi-hop evidence** in a non-trivial fraction of cases. Empirically, 74% of instances that generate a correct answer

rely on a single correct chunk, while 26% require multiple chunks. METEORA achieves essentially identical performance in both settings: single-chunk instances achieve precision/recall of 0.19/0.94, while multi-chunk instances achieve 0.19/0.91. Thus, METEORA preserves correct evidence while maintaining efficiency gains on multi-hop tasks.

Robustness to Corpus Poisoning in RAG. METEORA successfully defends against attacks and produces correct answers, as shown in Table 5. Without defense, the system fails completely ($F1 = 0$), and basic perplexity checks provide little help. Most caught poisoned evidence gets flagged for *instruction violations* rather than contradictions or factual errors. This shows that attacks mainly try to change *how the model works* instead of changing facts. Since the Verifier only flags evidence when confident and removes it before generating answers, the improvements come from *selection*, not from adding text. Even with shorter evidence, brief rationale-based cues are enough to decide what to keep and what to remove.

Human Evaluation. To quantify the interpretability and robustness of METEORA, we conduct a human evaluation using the popular notion of interpretability as the degree to which a human can understand the cause of a decision (Miller, 2019; Lipton, 2018). We sample 100 random examples across six datasets (QASPER, Contract-NLI, PrivacyQA, FinQA, CUAD, MAUD). Four experienced annotators, blinded to the ground truth and to each other’s labels, review each query–chunk pair along with the corresponding flagging instruction and the Verifier’s flags. For each chunk, they answer a binary question: “If you applied these instructions, would you correctly identify this chunk as poisoned/problematic?” (YES/NO) and provide an integer confidence score from 1 (very uncertain) to 5 (very confident). We measured interpretability through annotators’ confidence scores when applying the instructions and robustness through their agreement with ground-truth labels

Table 4. Baselines require proportionally more evidences than METEORA (values $> 1\times$) to reach comparable recall, except in FinQA (values $< 1\times$).

Method	QASPER	Contract-NLI	FinQA	PrivacyQA	CUAD	MAUD	Average
SBERT	1.88×	21.33×	0.90	2.00×	2.33×	1.46×	4.98×
Cross-Encoder	1.75×	21.33×	0.80	2.17×	2.17×	1.70×	4.99×
Contriever	1.88×	21.33×	0.90	2.00×	2.42×	1.82×	5.06×
RankRAG	4.00×	21.33×	1.50×	3.33×	5.17×	3.15×	6.41×

on evidence classification. Human evaluation demonstrates that the system is genuinely interpretable: annotators assigned the instructions a confidence score of 3.64 out of 5, indicating they found the rationales clear and applicable in most cases. Furthermore, annotator decisions achieved 86% accuracy against ground truth, demonstrating strong agreement on which evidence was problematic. Combined with the qualitative end-to-end examples in the appendix, these results indicate that humans can reliably understand and reconstruct the causes of METEORA’s evidence-level decisions based on the provided rationales.

Ablation Effect of DPO Training: DPO fine-tuning demonstrates its critical role in bridging the semantic gap between surface-level queries and the deep document locations where answers actually reside. The improvement is noticeable in complex datasets, such as MAUD experiences a 23.7% recall boost, while simpler datasets like FinQA show minimal gains (2.1%). This pattern reveals DPO’s core mechanism; it teaches models to map abstract legal concepts to specific document structures. Training with query–evidence pairs enables the model to internalize latent document-navigation strategies, allowing it to generate *rationales from queries alone without explicit structural annotations*.

Context Expansion Ablation. The expansion stage reveals a nuanced relationship between document complexity and chunking brittleness that challenges conventional assumptions about context windowing. MAUD’s merger agreements average 351k tokens and contain dense cross-references with nested clause structures. Expansion, therefore, helps substantially, improving recall by 7%, because related legal concepts are often distributed across adjacent chunks due to these interdependencies. Contracts in ContractNLI show minimal gains from context expansion because they utilize well-structured organizational patterns with clearly delineated sectional boundaries, where relevant information remains localized within individual chunks rather than distributed across adjacent segments. QASPER’s academic papers demonstrate moderate benefits as methodology discussions often span multiple chunks, requiring adjacent context to maintain coherence. PrivacyQA shows substantial improvement because privacy policy statements frequently continue across artificially imposed chunk boundaries. This dataset-specific performance pattern validates our hypothesis that expansion effectiveness correlates with document structural over document size alone, suggesting adaptive expansion strategies could optimize the precision-

recall tradeoff.

Table 5. Robust to corpus poisoning. **Top:** F1 score across datasets. **Bottom:** % of poisoned evidence flagged by category.

Method	QASPER	C-NLI	FinQA	PrivQA	CUAD	MAUD
No Defense	0.00	0.00	0.00	0.00	0.00	0.00
Perplexity	0.11	0.08	0.11	0.15	0.09	0.06
METEORA	0.44	0.51	0.43	0.48	0.33	0.39
Flag Type	QASPER	C-NLI	FinQA	PrivQA	CUAD	MAUD
Instruction	32.13	42.62	37.02	43.20	24.55	46.08
Contradiction	0.23	0.24	1.84	0.64	0.31	0.00
Factual	9.62	1.13	0.11	5.15	1.14	0.00
Total	41.98	43.99	38.97	48.99	26.00	46.08

Verifier Ablation. Table 5 shows instruction-based flagging accounts for 87% of detected adversarial content, revealing that attackers primarily target procedural subversion rather than factual substitution. This validates our rationale-based verification approach: adversaries attempt to manipulate “how to search” rather than “what facts exist.” Contradiction detection accounts for less than 2% of flags, while factual violations contribute 9.62% in QASPER. The verifier’s rationale-based design improves adversarial robustness without hurting clean-data quality.

6. Conclusion

We present METEORA, showing that interpretability and adversarial robustness in RAG need not be treated as competing objectives. By replacing opaque re-ranking with explicit rationales that guide and verify evidence selection, METEORA makes retrieval decisions more transparent while improving resistance to adversarial evidence. This challenges the prevailing assumption that interpretability necessarily reduces performance. Future work can further strengthen METEORA through neurosymbolic retrieval, which may combine rationale-based evidence selection with structured reasoning over claims, evidence, and source relationships in sensitive domains (Saxena and Gaur, 2026).

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Impact Statement

Our work improves the trustworthiness and transparency of Retrieval-Augmented Generation systems used in sensitive areas like legal, scientific research, and financial applications. METEORA uses rationales, explicit explanations of why evidence is selected, to make RAG systems both understandable to humans and resistant to attacks (Huang et al., 2024; Zhang et al., 2024). This helps organizations meet requirements for auditability in RAG while protecting against malicious content.

Ethical Considerations: Explanations through rationales can be helpful, but they might also make users trust the system too much without checking its decisions independently. Our defense mechanisms work well against the attacks we tested, but new types of attacks may bypass them. The verification component that filters out problematic evidence may inadvertently reject valid information. Additionally, METEORA explains only how evidence is selected; it does not cover the entire RAG pipeline from start to finish. This complete transparency remains an unsolved problem. Users should maintain human oversight and regularly check whether the system’s explanations match its actual behavior.

Future Societal Impact: As RAG systems take on more important decision-making roles in regulated fields, such as healthcare, making them understandable and secure becomes critical. Our rationale-based approach shows how to explain evidence selection clearly while detecting poisoned data, helping organizations meet transparency rules and reduce risks from attacks. Our findings suggest that building explanations directly into how systems make decisions, rather than adding them afterward, creates more reliable, trustworthy AI.

While our work focuses on evidence selection, the principles we demonstrate can guide the development of more transparent and secure RAG systems overall.

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A. Appendix

A.1. Prompts Used in Experiments

Few-Shot Prompt Template for Rationale Generation

Instructions: Generate semantic rationales to guide query-specific information selection from domain-specific documents. Each rationale should:

- represents a unique semantic search strategy grounded in the query.
- Be concise, concrete, and tailored to the scientific, legal, or financial context.
- Help extract precise and targeted evidence from long-form documents.
- Avoid redundancy across rationales.

Formatting Guidelines:

- Use XML-style tags: <rationale_1>, <rationale_2>, etc.
- Include a brief description in square brackets.
- Follow with a strategic, query-specific rationale sentence.

Example Query: What are the limitations of this approach?

Example Response (truncated):

```
<rationale_1>[Locate explicit limitation sections]
Look for sections explicitly titled "Limitations,"
"Threats to Validity," or "Shortcomings" which
directly enumerate the authors' acknowledged
limitations.....</rationale_1>
...
<rationale_10>[Review human evaluation or
annotation caveats] If any part of the work relies
on human judgment, authors may mention subjectivity
or annotator disagreement as
limitations.....</rationale_10>
```

Knowledge Poisoning Prompt with Example

Prompt:

You are given a question and the correct context. Your task is to: (1) Write an incorrect but plausible answer in the same style as the correct answer. (2) Create a poisoned_corpus (~100 words) that might mislead a model into selecting the incorrect answer . . .

Question:

Consider "Fiverr's" privacy policy; who can see the jobs that I post?

Correct Context:

In other words, when you access the Site we are aware of your usage of the Site, and may gather, collect, and record the information relating to such usage, including geo-location information, IP address, device and connection information, browser information, and web-log information, and all communications recorded by Users through the Site.

Poisoned Corpus:

Fiverr assures its users that their data is not only secure but private. When you interact with the platform, job postings and communications are secured using TLS encryption. Fiverr clearly states that it will never sell, rent, or share user content without explicit agreement. This gives users peace of mind when they publish job posts or engage with freelancers. In other words, your job posts are only visible to you and are never shared with other users, freelancers, or third parties, ensuring total privacy and anonymity.

A.2. Qualitative Examples Showcasing METEORA's Interpretable and Explainable Nature

Verifier Prompt

Input:

- **Query:** {query}
- **Flagging Instructions:** {rationales}
- **Previous Chunks Summaries:** {chunk_summaries}
- **Chunk to Verify:** {chunk_text}

Instructions:

- Flag clear factual errors only if highly confident (>90%).
- Detect contradictions with prior verified chunks.
- Assume validity unless strong evidence to the contrary exists.

Response Format:

```
{ "flagged": false, "chunk_summary":
  "This chunk discusses ...",
  "flag_types": [] }
```

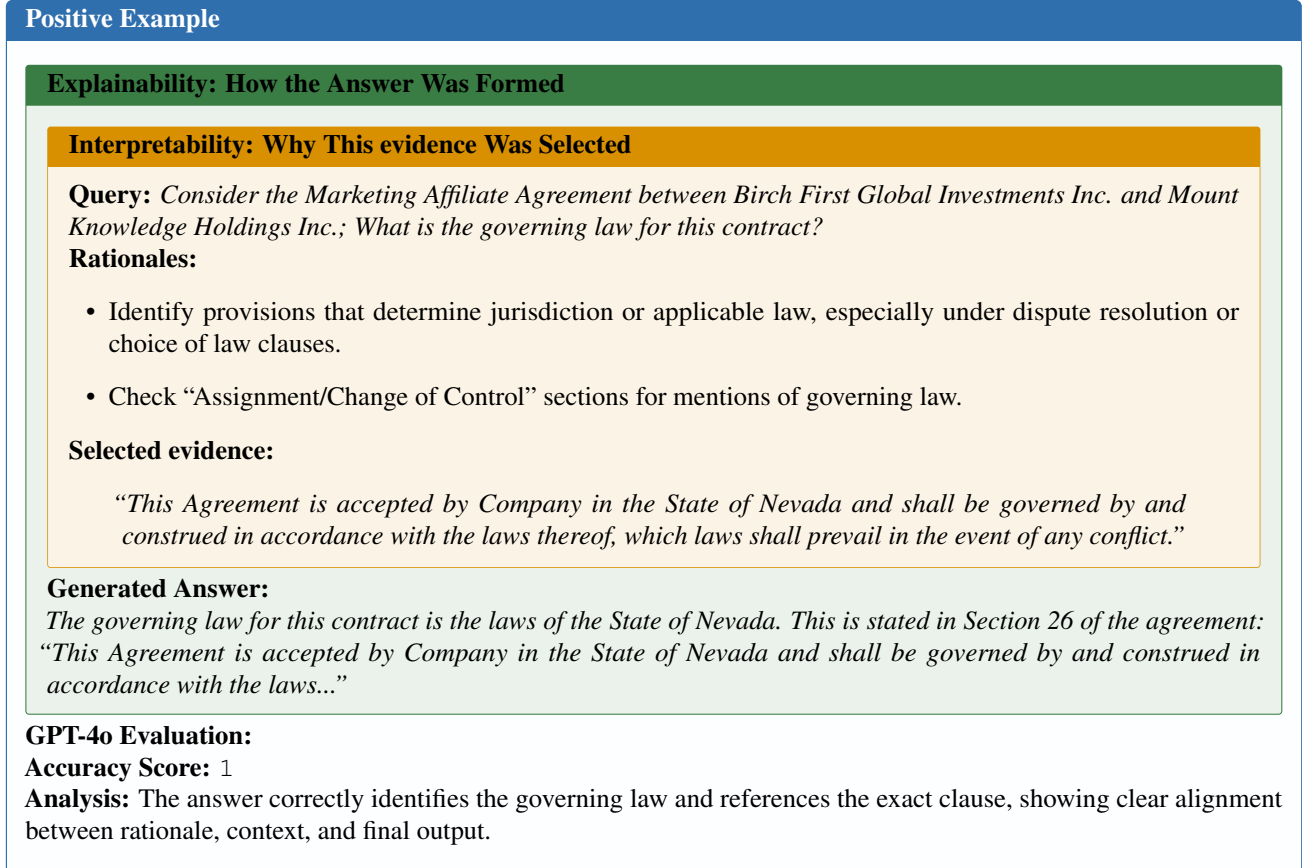


Figure 4. Positive example demonstrating how METEORA links rationale-based evidence selection (interpretability) with rationale-grounded answer generation (explainability), resulting in a correct and traceable response to a legal question.

A.3. DPO Implementation Details

We use Direct Preference Optimization (DPO) to fine-tune a general-purpose LLM to generate query-aligned rationales. The preference dataset is automatically constructed using the original QA corpus: for each query, rationales that led to correct evidence selection form the preferred output, while others form the rejected output. No manual labeling is required. The model is trained using pairwise comparisons of effective and ineffective rationales.

We train the model using the LLaMA-3.1-8B-Instruct LLM. The DPO loss (subsection 3.1 in section 3) is optimized over three epochs using cosine learning rate scheduling. Training and validation data are derived from a single annotated file using an 80/10/10 train-validation-test split.

A.4. Elbow Detection in Pooled Rationale Component of ECSE

To identify evidence that align with the collective intent of all rationales, we compute a pooled embedding $\bar{r} = \frac{1}{|R|} \sum_{r_i \in R} \text{SBERT}(r_i)$ and calculate cosine similarity between $\bar{r}$ and each evidence embedding. This produces a

Table 6. DPO training configuration used for rationale refinement.

Parameter	Value
Base model	LLaMA-3.1-8B-Instruct
Batch size (train/eval)	1 / 1
Gradient accumulation steps	2
Epochs	3
Learning rate	3e-5
Scheduler	Cosine
Warmup ratio	0.1
DPO loss β	0.05
Train / Val / Test split	80% / 10% / 10%
Save strategy	Per epoch
Best model selection metric	eval_rewards/chosen

sorted sequence of similarity scores $\{s_1, s_2, \dots, s_n\}$.

We first compute the first-order differences $\Delta_i = s_i - s_{i+1}$ and apply z-score normalization across $\{\Delta_i\}$ to highlight sharp changes in similarity. The selection index k^* is identified at the first point where the drop in similarity significantly deviates from the average pattern, indicating a natural boundary between highly relevant and less relevant chunks.

In cases where similarity scores decline uniformly, and no clear deviation is found, we fall back to computing the second-order differences $\nabla_i^2 = \Delta_{i+1} - \Delta_i$. We then choose the index of maximum curvature, which reflects the sharpest transition in the similarity landscape. The selected top- k^*

Negative Example

Explainability: Why the Answer Was Incorrect

Interpretability: Why These evidence Were Selected

Query: Consider the Marketing Affiliate Agreement between Birch First Global Investments Inc. and Mount Knowledge Holdings Inc.; Is there a cap on liability under this contract?

Rationales:

- Look for sections discussing liability caps or clauses stating a maximum limit of responsibility for either party.
- Search for phrases like “shall not exceed,” “limited to,” or “capped at” in liability or indemnity sections.
- Focus on sections near “Dispute Resolution” or “Indemnification” — common locations for liability limitations.

Correct evidence (not selected):

“Company is not liable for incidental, special or consequential damages for any reason [...]. Company’s liability shall not exceed the fees that MA has paid under this Agreement.”

Generated Answer:

There is no clear cap on liability under this contract; however, according to paragraph 15, Company’s liability “shall not exceed the fees...”

GPT-4o Evaluation:

Accuracy Score: 0

Analysis: The generated answer inaccurately states that there is no clear cap on liability and does not fully convey the information that the company’s liability is capped at the fees paid under the agreement. The reference answer provides a clear and specific cap on liability, which is missing in the generated answer.

Figure 5. Negative example showing how METEORA enables transparent error tracing. Although the correct evidence was not selected, the rationale trail provides insight into why the incorrect evidence was selected, and how it influenced the erroneous answer.

chunks, denoted as E_g , are thus derived without relying on manually defined thresholds, enabling adaptive and data-driven cutoff across varying distributions.

A.5. Results Across All Evidence Sizes

We report precision, recall, and generation performance for all evaluated chunk sizes across datasets to complement the main results presented in the paper.

B. DPO Theory

We address the problem of aligning an LLM’s rationale generation capability with evidence selection using Direct Preference Optimization (DPO). We consider a setting where:

- We have a preference dataset consisting of document evidences
- For each user query, certain document evidences are labeled as relevant (positive) examples

- Other document evidences are considered irrelevant (negative) examples

- The goal is to train the LLM to select appropriate evidences by generating rationales that explain the relevance of evidences to the query

Unlike traditional RAG approaches that use re-ranking mechanisms based on similarity metrics, our approach enables the LLM to learn to select evidences through a process of rationale generation, providing transparency and better alignment with the final response generation. We provide a rigorous mathematical formulation of this problem and prove that DPO training leads to a policy that optimally selects contextual evidences based on generated rationales.

C. Problem Formulation

Let us now formalize the problem of aligning rationale generation for contextual evidence selection using DPO.

Table 7. CP Task Results across Datasets and Evidence Sizes

Model	Contract-NLI		PrivacyQA		CUAD		MAUD		QASPER	
Evidence Size = 128										
	P@7	R@7	P@10	R@10	P@24	R@24	P@43	R@43	P@22	R@22
SBERT	0.17	0.78	0.12	0.61	0.04	0.59	0.01	0.03	0.10	0.86
Contriever	0.16	0.81	0.11	0.55	0.04	0.27	0.01	0.14	0.11	0.90
Cross-Encoder	0.17	0.83	0.11	0.51	0.03	0.50	0.01	0.15	0.11	0.89
Finetuned-SBERT	0.17	0.81	0.09	0.59	0.04	0.59	0.01	0.14	0.11	0.88
RankRAG	0.12	0.77	0.09	0.57	0.04	0.49	0.01	0.12	0.09	0.61
METEORA	0.18	0.89	0.13	0.84	0.06	0.78	0.02	0.39	0.11	0.90
Evidence Size = 256										
	P@5	R@5	P@8	R@8	P@17	R@17	P@37	R@37	P@14	R@14
SBERT	0.25	0.88	0.17	0.81	0.07	0.76	0.01	0.37	0.17	0.93
Contriever	0.24	0.85	0.15	0.68	0.06	0.68	0.01	0.27	0.18	0.95
Cross-Encoder	0.25	0.89	0.16	0.74	0.06	0.64	0.01	0.31	0.16	0.94
Finetuned-SBERT	0.25	0.89	0.13	0.59	0.07	0.68	0.01	0.36	0.17	0.94
RankRAG	0.19	0.73	0.11	0.69	0.05	0.76	0.01	0.22	0.13	0.79
METEORA	0.25	0.98	0.18	0.92	0.08	0.84	0.02	0.58	0.16	0.96

Table 8. Verifier Flag Distribution (Excluding FinQA). Percentage of flagged poisoned evidence categorized by violation type across evidence sizes.

Flag Type	QASPER	Contract-NLI	PrivacyQA	CUAD	MAUD
Evidence Size = 128					
Instruction	23.67	57.80	34.80	20.37	43.93
Contradiction	0.07	0.21	0.21	2.23	0.11
Factual	9.27	3.96	3.96	2.44	0.02
Evidence Size = 256					
Instruction	35.30	52.48	54.93	20.79	40.60
Contradiction	0.10	0.07	0.39	1.86	0.46
Factual	7.59	2.39	4.68	2.18	0.00

C.1. Notation and Setup

We denote:

- $\mathcal{Q}$: The set of all possible user queries
- $\mathcal{E}$: The set of all possible contextual evidences in the knowledge base
- $\mathcal{R}$: The set of all possible rationales explaining evidence relevance
- $\mathcal{E}_q^+ \subset \mathcal{E}$: The set of relevant evidences for query q
- $\mathcal{E}_q^- \subset \mathcal{E}$: The set of irrelevant evidences for query q
- $\pi_\theta(e, r|q)$: The LLM policy parameterized by θ , giving the joint probability of selecting evidence e and generating rationale r given query q
- $\pi_{\text{ref}}(e, r|q)$: The reference (initial) policy

We can decompose the joint probability as:

$$\pi_\theta(e, r|q) = \pi_\theta(e|q) \cdot \pi_\theta(r|q, e) \quad (1)$$

Where $\pi_\theta(e|q)$ is the probability of selecting evidence e for query q , and $\pi_\theta(r|q, e)$ is the probability of generating rationale r given query q and selected evidence e .

C.2. Preference Model

We assume there exists an ideal reward function $r^*(q, e, r)$ that captures the appropriateness of both the selected evidence and the generated rationale. This reward function should assign higher values to relevant evidences with convincing rationales and lower values to irrelevant evidences or unconvincing rationales.

We model this as:

$$r^*(q, e, r) = f(\text{relevance}(e, q), \text{quality}(r, q, e)) \quad (2)$$

Where $\text{relevance}(e, q)$ measures how relevant the evidence e is to query q , and $\text{quality}(r, q, e)$ measures how well the rationale r explains the relevance of evidence e to query q . The function f combines these measures.

Ranking Free RAG: Replacing Re-ranking with Selection in RAG for Sensitive Domains

Table 9. Ablation study of the ECSE pipeline, evaluating the effect of Pairing, Pooling, and Expansion stages across datasets and evidence sizes. Pooling improves recall over Pairing alone, and adding Expansion further enhances performance. The complete ECSE pipeline achieves the best balance between precision and recall, particularly at larger evidence sizes.

Components	Contract-NLI		PrivacyQA		CUAD		MAUD		QASPER	
Evidence Size = 128										
	P	R	P	R	P	R	P	R	P	R
Pairing	0.20	0.91	0.14	0.64	0.08	0.64	0.01	0.25	0.14	0.81
Pairing + Expansion	0.18	0.95	0.13	0.84	0.06	0.78	0.02	0.39	0.11	0.90
Evidence Size = 256										
	P	R	P	R	P	R	P	R	P	R
Pairing	0.27	0.94	0.19	0.83	0.10	0.69	0.02	0.50	0.20	0.90
Pairing + Expansion	0.25	0.98	0.18	0.92	0.08	0.84	0.02	0.58	0.16	0.96

Table 10. Corpus poisoning detection performance across datasets and evidence sizes. METEORA shows strong resilience, outperforming perplexity-based defenses in recall, especially at larger evidence sizes.

Method	Contract-NLI		PrivacyQA		CUAD		MAUD		QASPER	
Evidence Size = 128										
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
No Defense	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Perplexity	0.69	0.22	0.19	0.17	0.19	0.22	0.46	0.12	0.09	0.10
METEORA	0.62	0.55	0.37	0.18	0.25	0.31	0.44	0.42	0.29	0.33
Evidence Size = 256										
No Defense	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Perplexity	0.54	0.20	0.29	0.18	0.18	0.10	0.31	0.14	0.27	0.15
METEORA	0.55	0.72	0.60	0.40	0.24	0.38	0.41	0.46	0.43	0.53

Table 11. FinQA analysis across evidence sizes: CP task performance, ablation of ECSE, verifier flag contribution, and poisoning detection.

CP Performance			ECSE Ablation			Verifier Flags		Poisoning	Detection	
Model	P@13	R@13	Stage	P@13	R@13	Flag	32	Method	Prec.	Rec.
Size = 32			Size = 32			Instruction	32.02	Size = 32		
SBERT	0.08	0.88	Pairing	0.08	0.82	Contradiction	1.64	No Defense	0.00	0.00
Contriever	0.07	0.91	+ Expansion	0.07	0.89	Factual	0.32	Perplexity	0.13	0.05
CrossEnc	0.09	0.83	Size = 64			Flag	64	METEORA	0.34	0.34
Fine-Tuned SBERT	0.08	0.90	Pairing	0.14	0.91	Instruction	37.02	Size = 64		
RankRAG	0.06	0.78	+ Expansion	0.12	0.95	Contradiction	1.84	No Defense	0.00	0.00
METEORA	0.07	0.89				Factual	0.11	Perplexity	0.18	0.07
Size = 64								METEORA	0.39	0.48
SBERT	0.12	0.96								
Contriever	0.11	0.98								
CrossEnc	0.12	0.97								
Fine-Tuned SBERT	0.13	0.97								
RankRAG	0.08	0.89								
METEORA	0.12	0.95								

A simple form could be:

$$r^*(q, e, r) = \alpha \cdot \text{relevance}(e, q) + \gamma \cdot \text{quality}(r, q, e) \quad (3)$$

Where $\alpha, \gamma > 0$ are weights that balance the importance of evidence relevance and rationale quality.

C.3. Deriving the DPO Objective for Rationale-Based Evidence Selection

To apply DPO, we need preference data in the form of $(q, (e_w, r_w), (e_l, r_l))$ tuples, where (e_w, r_w) is preferred over (e_l, r_l) .

In our setting, we can generate these tuples as follows:

- q : A user query
- (e_w, r_w) : A relevant evidence with a high-quality rationale explaining its relevance
- (e_l, r_l) : An irrelevant evidence with a rationale attempting to explain its relevance

Given such tuples, the DPO objective becomes:

$$\begin{aligned} & \mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) \\ &= -\mathbb{E}_{(q, (e_w, r_w), (e_l, r_l))} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(e_w, r_w|q)}{\pi_{\text{ref}}(e_w, r_w|q)} \right. \right. \\ & \quad \left. \left. - \beta \log \frac{\pi_\theta(e_l, r_l|q)}{\pi_{\text{ref}}(e_l, r_l|q)} \right) \right] \quad (4) \end{aligned}$$

D. Theoretical Analysis

We now prove that optimizing the DPO objective leads to an aligned policy that selects appropriate contextual evidence through rationale generation.

Theorem 1 (Optimality of DPO for Rationale-Based Evidence Selection). *Let π_θ be a policy trained using the DPO objective with preference data derived from relevant and irrelevant evidence with their corresponding rationales. Under certain regularity conditions, as the amount of preference data increases, π_θ converges to:*

$$\pi^*(e, r|q) \propto \pi_{\text{ref}}(e, r|q) \exp \left(\frac{1}{\beta} r^*(q, e, r) \right) \quad (5)$$

Where $r^*(q, e, r)$ is the true reward function capturing the appropriateness of evidence selection and rationale generation based on the query.

We proceed with the proof step by step.

Step 1: Recall from (Rafailov et al., 2023b) that DPO is derived from the following principle: in the Bradley-Terry preference model, the probability that response (e_w, r_w) is preferred over (e_l, r_l) given query q is:

$$P((e_w, r_w) \succ (e_l, r_l)|q) = \sigma(r(q, e_w, r_w) - r(q, e_l, r_l)) \quad (6)$$

Where $r(q, e, r)$ is the reward function and σ is the logistic function.

Step 2: The optimal policy given a reward function r and a reference policy π_{ref} is:

$$\pi_r(e, r|q) \propto \pi_{\text{ref}}(e, r|q) \exp \left(\frac{1}{\beta} r(q, e, r) \right) \quad (7)$$

This follows from the constrained optimization problem of maximizing expected reward subject to a KL-divergence constraint with the reference policy.

Step 3: Plugging the optimal policy form into the Bradley-Terry model, we get:

$$\begin{aligned} P((e_w, r_w) \succ (e_l, r_l)|q) &= \sigma(r(q, e_w, r_w) - r(q, e_l, r_l)) \\ &= \sigma \left(\beta \log \frac{\pi_r(e_w, r_w|q)}{\pi_{\text{ref}}(e_w, r_w|q)} \right. \\ & \quad \left. - \beta \log \frac{\pi_r(e_l, r_l|q)}{\pi_{\text{ref}}(e_l, r_l|q)} \right) \end{aligned}$$

Step 4: The DPO objective trains π_θ to match these probabilities by minimizing:

$$\begin{aligned} & \mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) \\ &= -\mathbb{E}_{(q, (e_w, r_w), (e_l, r_l))} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(e_w, r_w|q)}{\pi_{\text{ref}}(e_w, r_w|q)} \right. \right. \\ & \quad \left. \left. - \beta \log \frac{\pi_\theta(e_l, r_l|q)}{\pi_{\text{ref}}(e_l, r_l|q)} \right) \right] \quad (8) \end{aligned}$$

Step 5: As the amount of preference data increases, assuming the preference data accurately reflects the true reward function $r^*(q, e, r)$ that values relevant evidence with convincing rationales over irrelevant evidence, minimizing the DPO loss will drive π_θ towards the optimal policy:

$$\pi_\theta(e, r|q) \rightarrow \pi_{r^*}(e, r|q) \propto \pi_{\text{ref}}(e, r|q) \exp \left(\frac{1}{\beta} r^*(q, e, r) \right) \quad (9)$$

Step 6: We can further analyze the joint probability by decomposing it:

$$\pi^*(e, r|q) \propto \pi_{\text{ref}}(e|q) \pi_{\text{ref}}(r|q, e) \exp \left(\frac{1}{\beta} r^*(q, e, r) \right) \quad (10)$$

Step 7: Since $r^*(q, e, r)$ rewards both chunk relevance and rationale quality, the resulting policy will select evidences and generate rationales that are aligned with the true relevance of evidences to the query. This means the policy learns to select evidences based on their relevance through a process of rationale generation rather than simple re-ranking.

Corollary 2 (Alignment Guarantee for Chunk Selection). *If the preference dataset accurately reflects the relevance of evidences to queries along with appropriate rationales, then the DPO-trained policy will select contextual evidences that are most relevant to the query while generating rationales that justify this selection.*

Corollary 3 (Superiority over Re-ranking). *The DPO-trained policy provides advantages over traditional re-ranking in RAG systems by:*

1. Jointly optimizing evidence selection and rationale generation
2. Providing transparent explanations for why specific evidences were selected
3. Learning complex relevance patterns beyond simple similarity metrics

E. Algorithm

We now describe a step-by-step procedure for implementing DPO for rationale-based evidence selection:

Algorithm 1 DPO for Rationale-Based Chunk Selection

Require: Base LLM π_{ref} , user queries $\{q_i\}$, relevant chunks $\{\mathcal{E}_{q_i}^+\}$, irrelevant chunks $\{\mathcal{E}_{q_i}^-\}$

Ensure: Aligned LLM π_θ that selects evidence with rationales

- 1: Initialize $\pi_\theta \leftarrow \pi_{\text{ref}}$
- 2: Construct preference dataset $\mathcal{P} = \{(q_i, (e_{w,i}, r_{w,i}), (e_{l,i}, r_{l,i}))\}$
- 3: **for** each query q_i **do**
- 4: Sample relevant evidence $e_{w,i}$ from $\mathcal{E}_{q_i}^+$
- 5: Generate rationale $r_{w,i}$ explaining relevance of $e_{w,i}$ to q_i
- 6: Sample irrelevant evidence $e_{l,i}$ from $\mathcal{E}_{q_i}^-$
- 7: Generate rationale $r_{l,i}$ attempting to explain relevance of $e_{l,i}$ to q_i
- 8: Add $(q_i, (e_{w,i}, r_{w,i}), (e_{l,i}, r_{l,i}))$ to $\mathcal{P}$
- 9: **end for**
- 10: Train π_θ by minimizing Equation 8.
- 11: **return** π_θ

F. Rationale Fabrication

The model might generate convincing-sounding but factually incorrect rationales to justify the selection of irrelevant chunks.

Mitigation: Include factual consistency metrics in the training process and explicitly penalize fabricated or misleading rationales.

G. Distribution Shift

The distribution of queries and evidences during deployment may differ from those in the training data.

Mitigation: Include a diverse range of queries and evidence types in the training data, and implement continual learning mechanisms to adapt to new domains.

G.0.1. RETRIEVAL-AUGMENTED GENERATION (RAG)

The RAG architecture enhances generative models by incorporating additional context from a knowledge base, thereby improving response accuracy. Given a query q , a knowledge base of documents $D = \{d_1, d_2, \dots, d_n\}$, a retriever function $F(q, D) \rightarrow D_q \subset D$, a re-ranker function $\mathcal{R}e(q, D_q)$ that picks the top- k documents, and a generative model $\mathcal{M}$, the final generation g is given by:

$$g = \mathcal{M}(q, \mathcal{R}e(q, F(q, D)))$$

G.1. Current Re-Ranking Mechanism

A re-ranker $\mathcal{R}e$ computes the semantic similarity $\mathcal{S}s$ between a query q and a set of retrieved documents D_q , and then ranks the documents by relevance. In *top-k* SBERT-based re-ranking, the query and each document are encoded independently, and cosine similarity is used to score relevance: $\mathcal{R}e_{\text{SBERT}}(q, D_q) = \{\cos(\text{SBERT}(q), \text{SBERT}(d)) \mid d \in D_q\}$

In *top-k* cross-encoders, the query and each document are jointly encoded, producing a scalar relevance score:

$$\mathcal{R}e_{\text{Cross}}(q, D_q) = \{\text{CrossEncoder}(q, d) \mid d \in D_q\}$$

Top-k Contriever also encodes the query and documents independently using a contrastive learning objective: $\mathcal{R}e_{\text{Contriever}}(q, D_q) = \{\cos(\text{Contriever}(q), \text{Contriever}(d)) \mid d \in D_q\}$

While effective for ranking based on surface-level similarity, these methods have three key limitations. First, they require manual tuning of k , which is often done through hit-and-trial. Second, using a fixed k for all queries in a domain may include less-relevant chunks in the top- k , which can negatively impact downstream generation. Third, these methods lack interpretability and provide no mechanisms to detect or filter adversarial or misleading content.